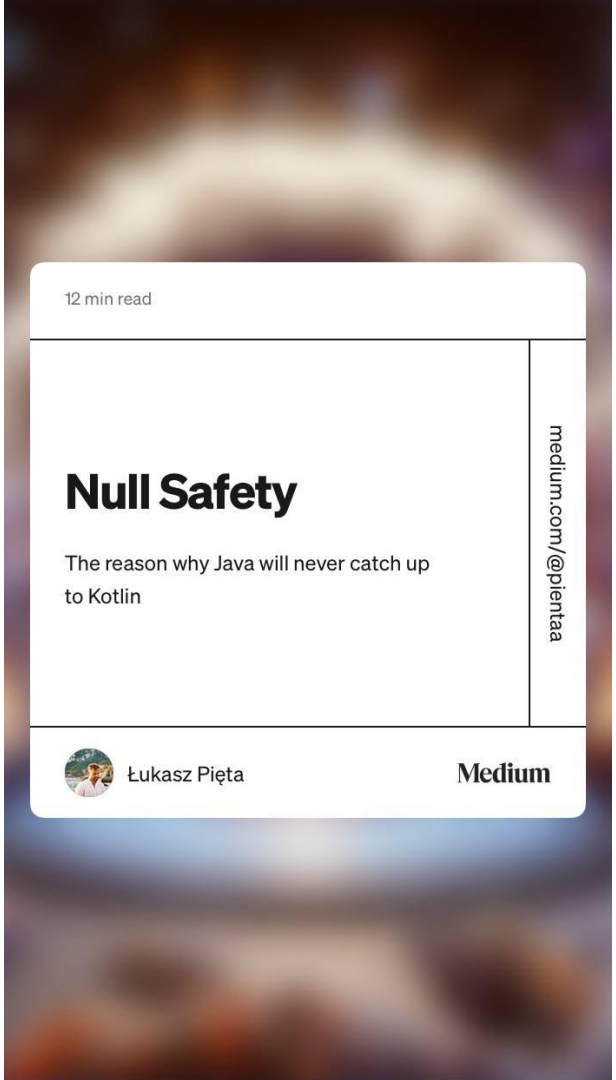
Embracing Clean Architecture

in Spring Boot with Kotlin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Software architecture

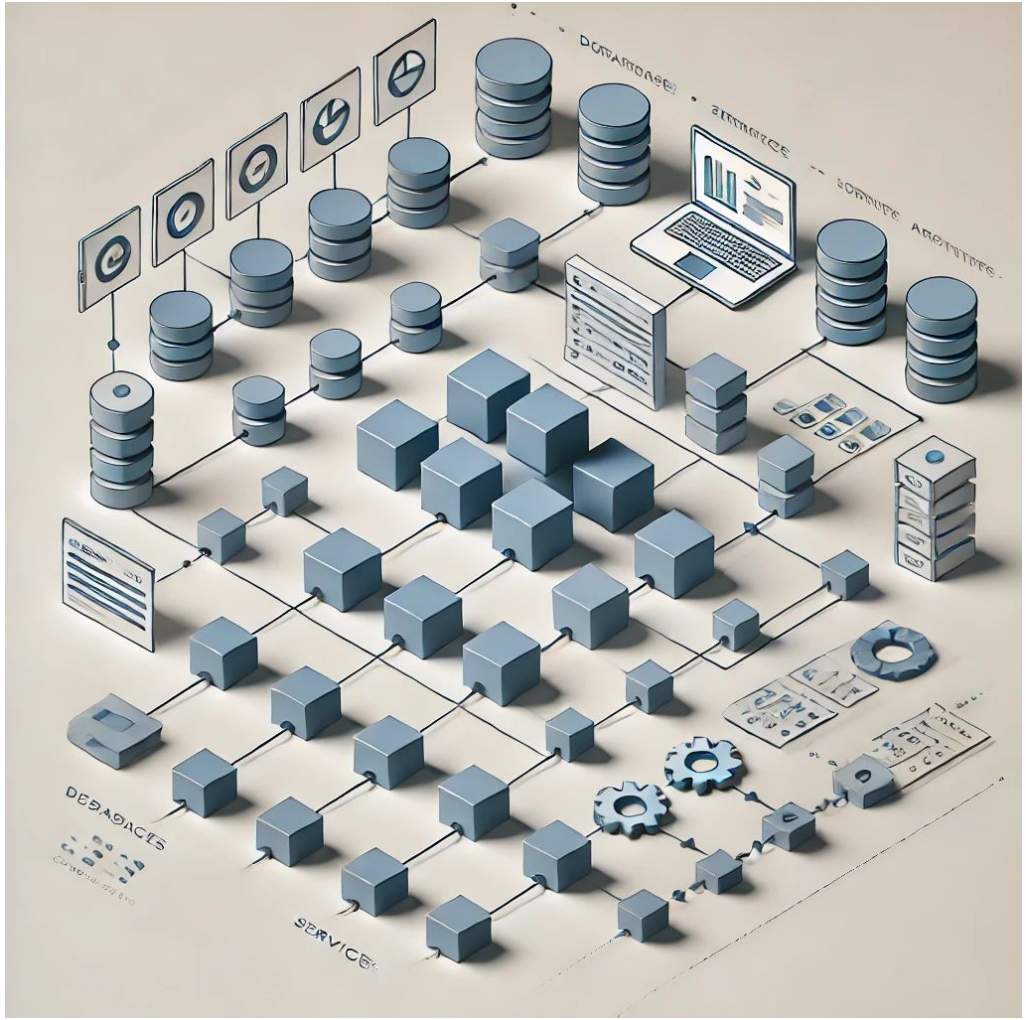
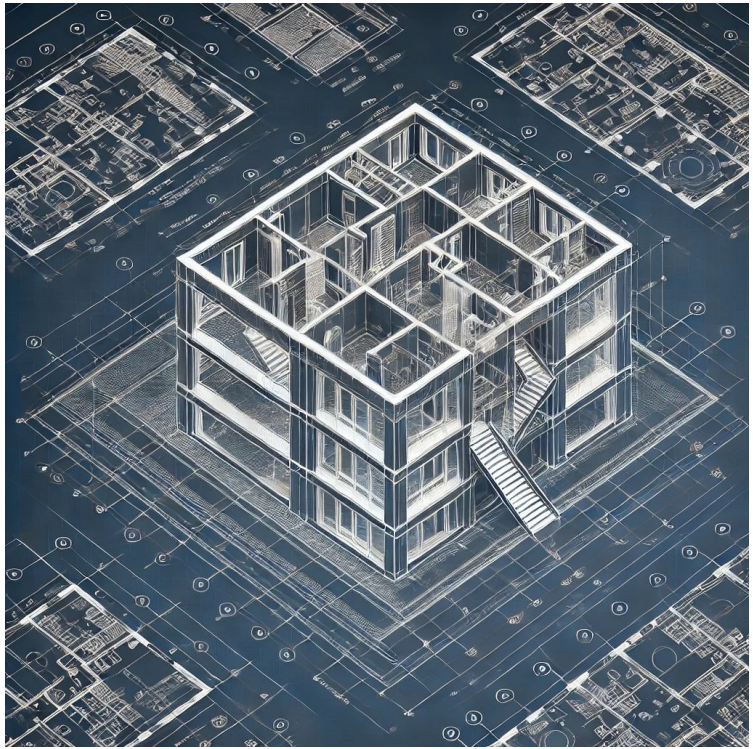


Image generated by AI using OpenAI's DALL-E

“Architecture” metaphor



Images generated by AI using OpenAI's DALL-E

“The set of **significant decisions** about the **organization of a software system**, including the choice of structural elements and their interfaces.”

~ Martin Fowler

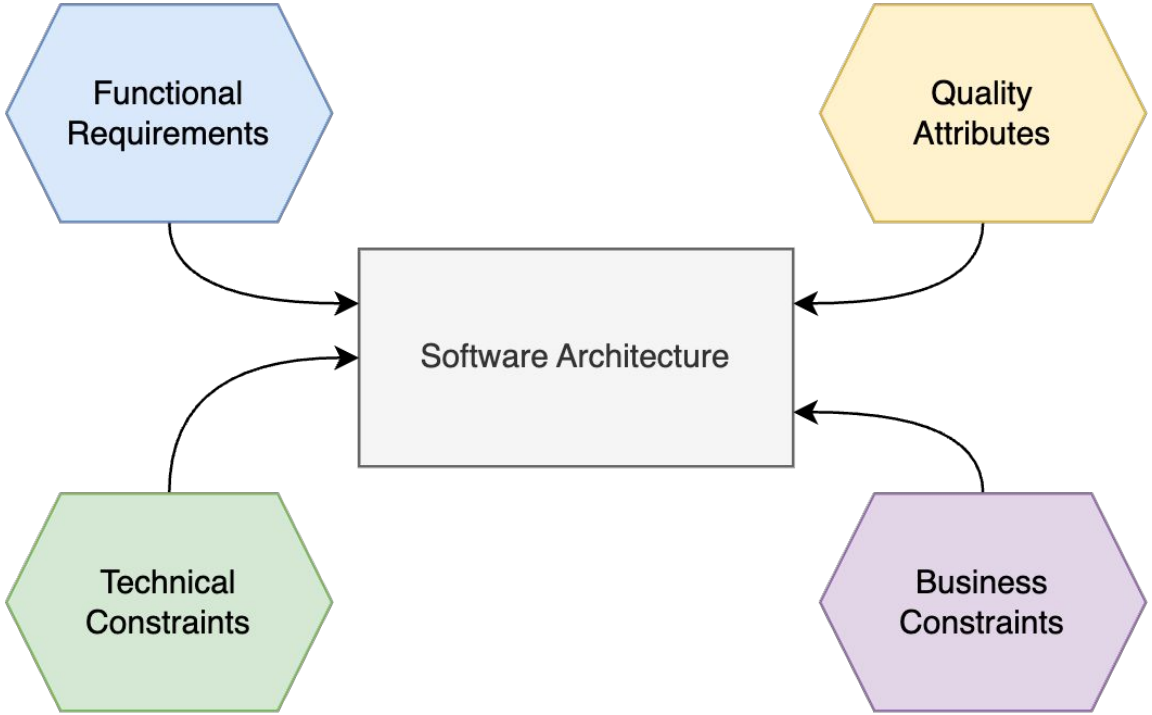
“Architecture is about **the important stuff**. Whatever that is.”

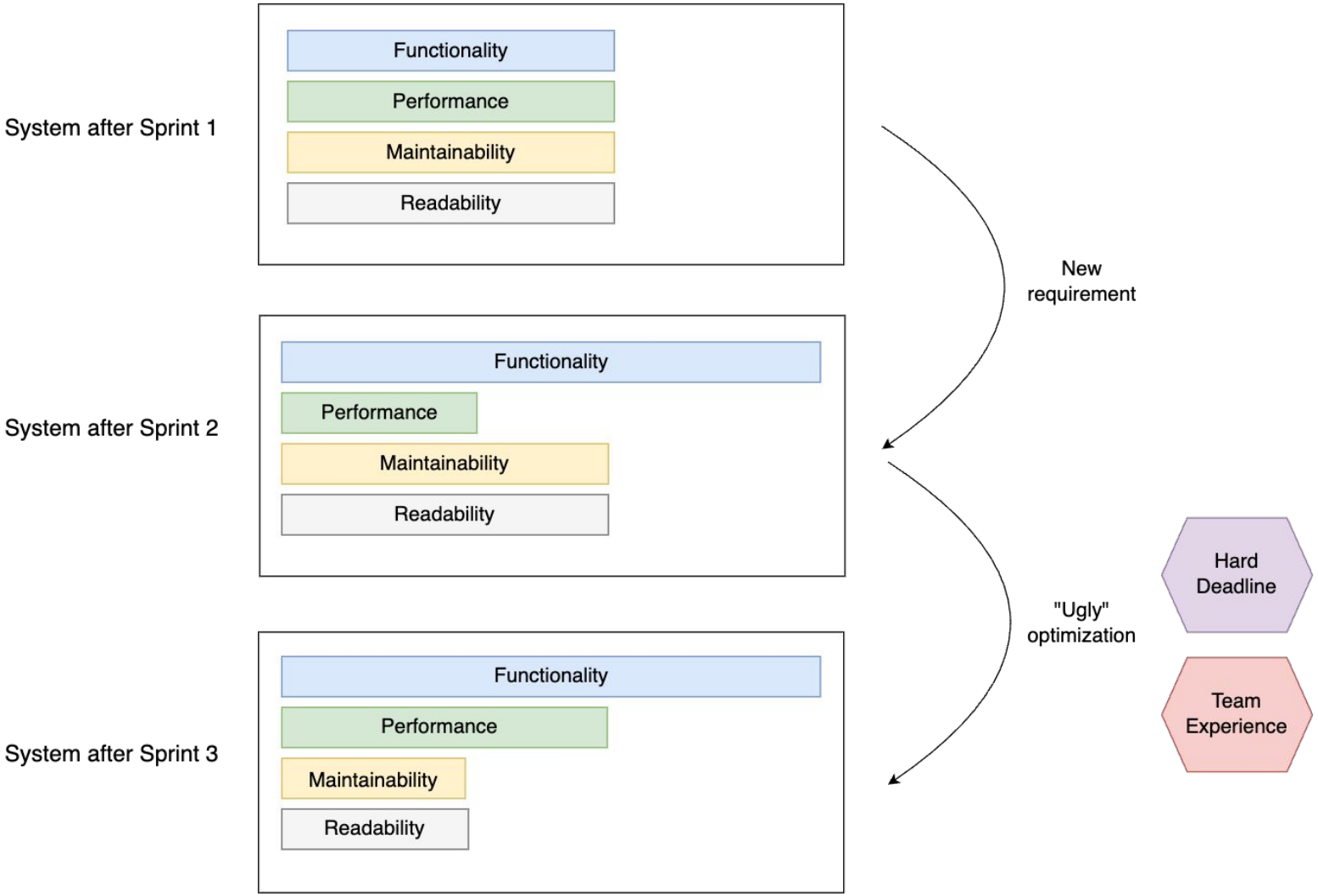
~ Ralph Johnson

Architectural Drivers

"Set of common things that really drive, influence and shape the resulting software architecture"

~ Simon Brown




System after Sprint 1
Functionality
Performance
Maintainability
Readability
New requirement
System after Sprint 2
Functionality
Performance
Maintainability
Readability
"Ugly" optimization
Hard Deadline
Team Experience
System after Sprint 3
Functionality
Performance
Maintainability
Readability

TRADE-OFFS
LIGHTYEAR
TRADE-OFFS EVERYWHERE
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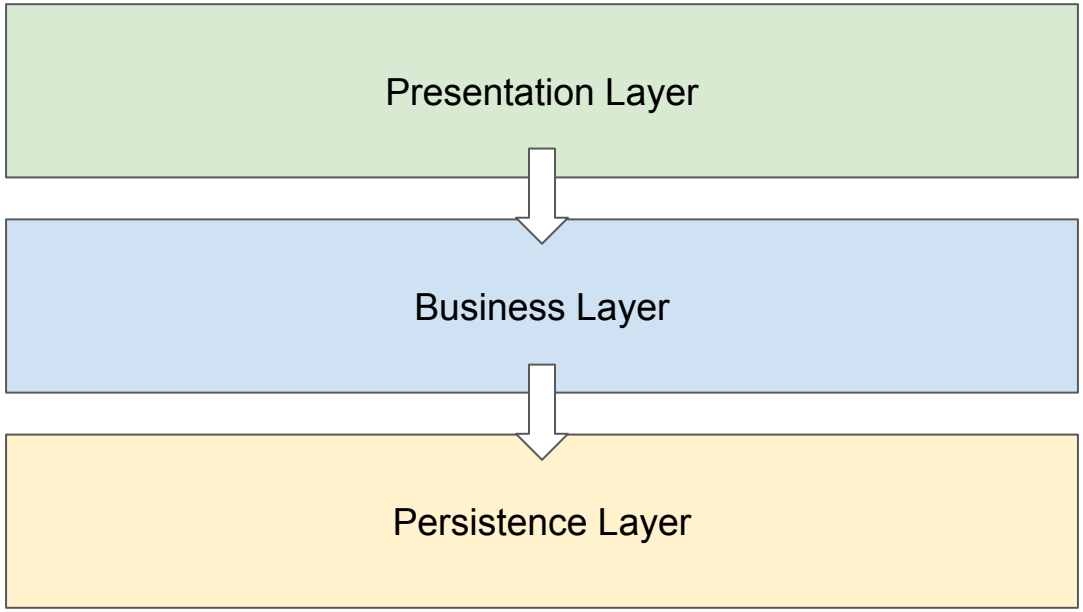
Most Common Architectural Styles

Layered architecture



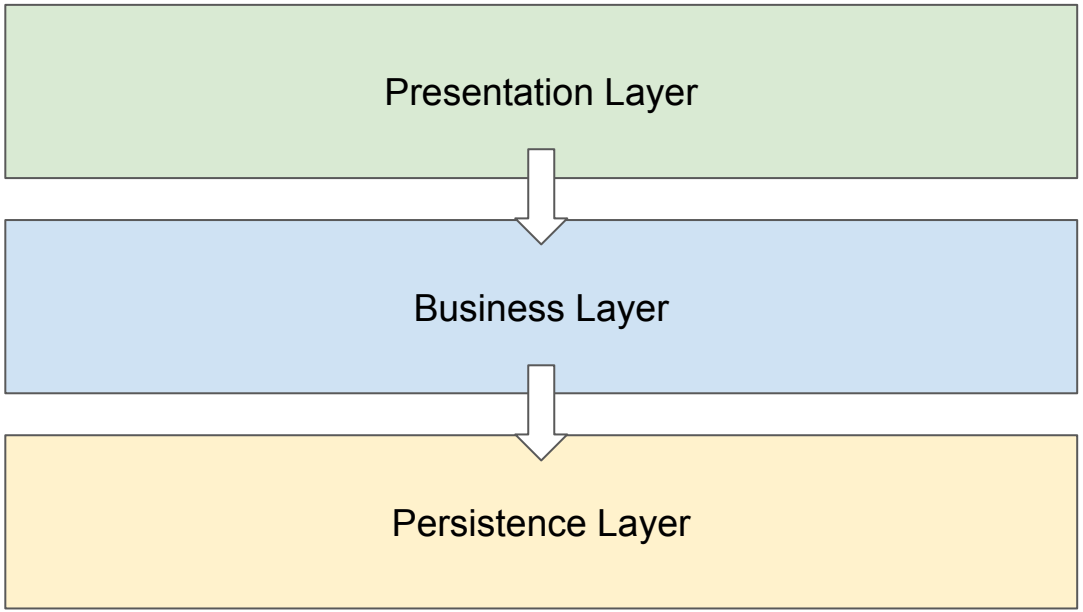
Layered architecture

Pros:

- Just simple

Cons:

- Tight coupling
- Maintainability rapidly decreasing with increasing complexity



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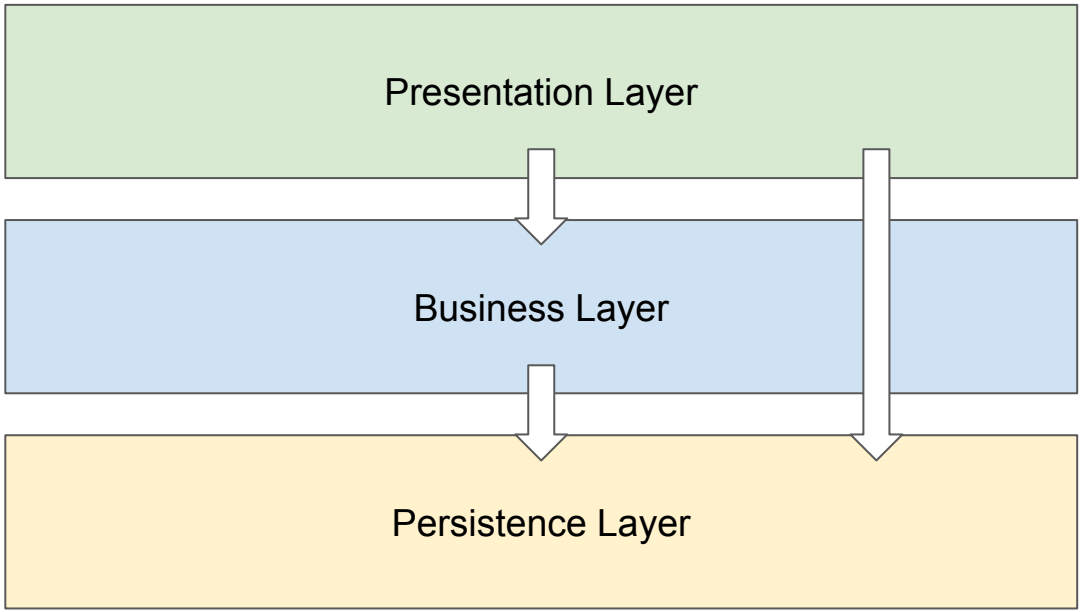


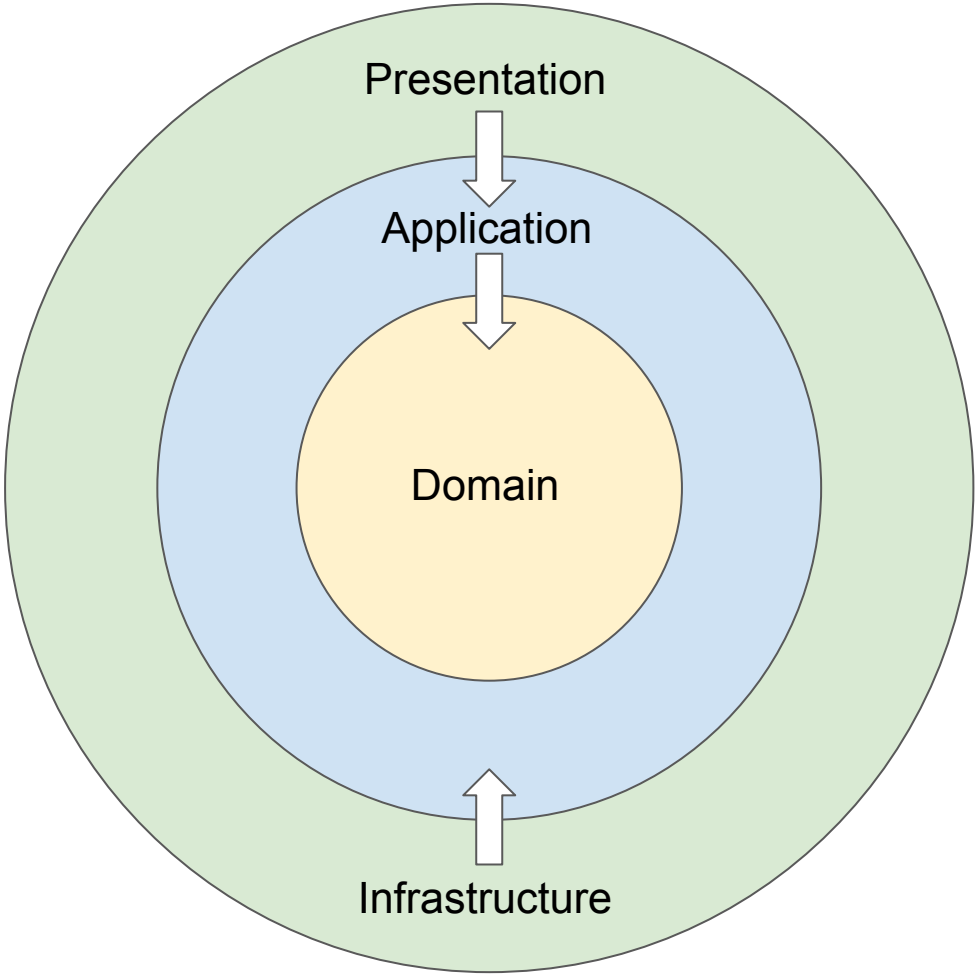
Onion architecture

Pros:

- Testability
- Separation of Concerns
- Flexibility

Cons:

- Initial Complexity
- Learning Curve
- Increased Boilerplate Code



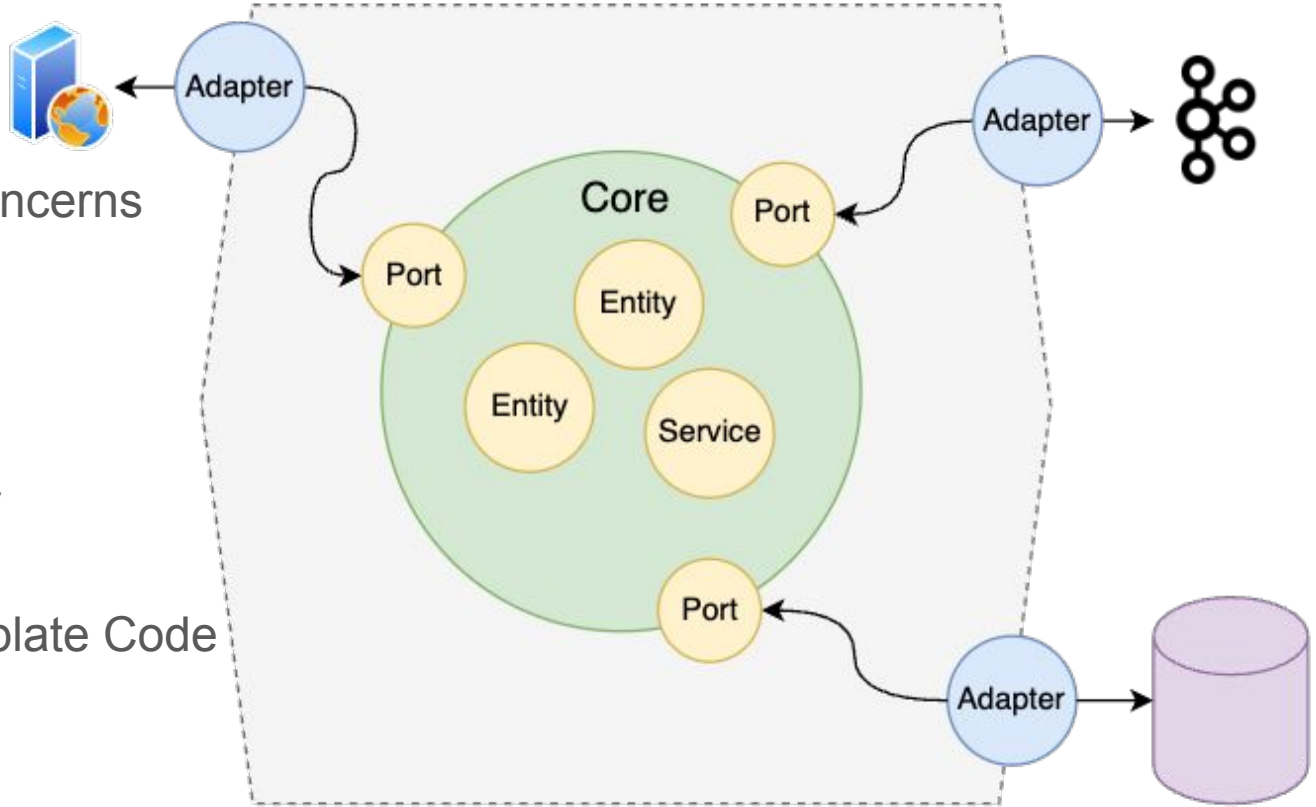
Hexagonal architecture

Pros:

- Testability
- Separation of Concerns
- Flexibility

Cons:

- Initial Complexity
- Learning Curve
- Increased Boilerplate Code



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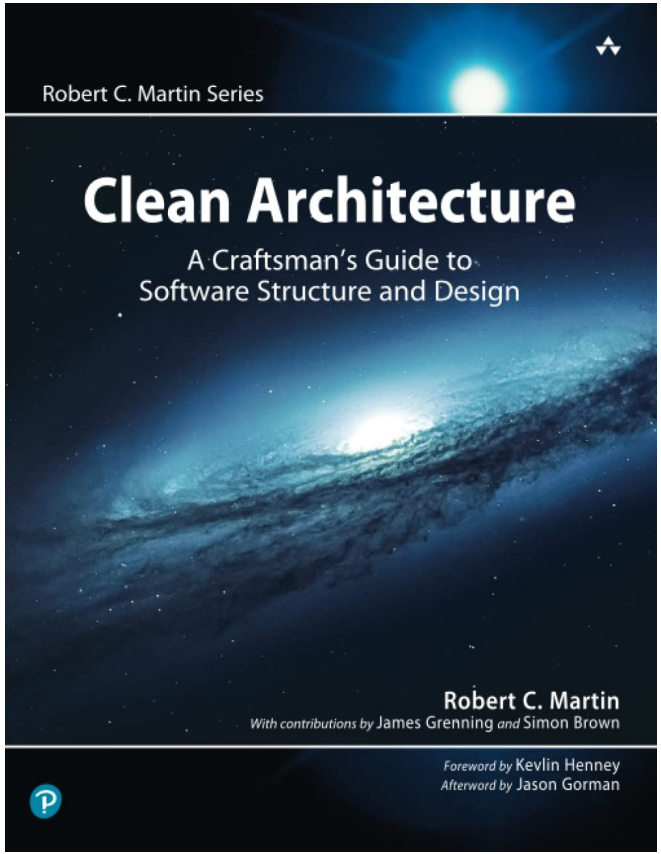


“Clean architecture”

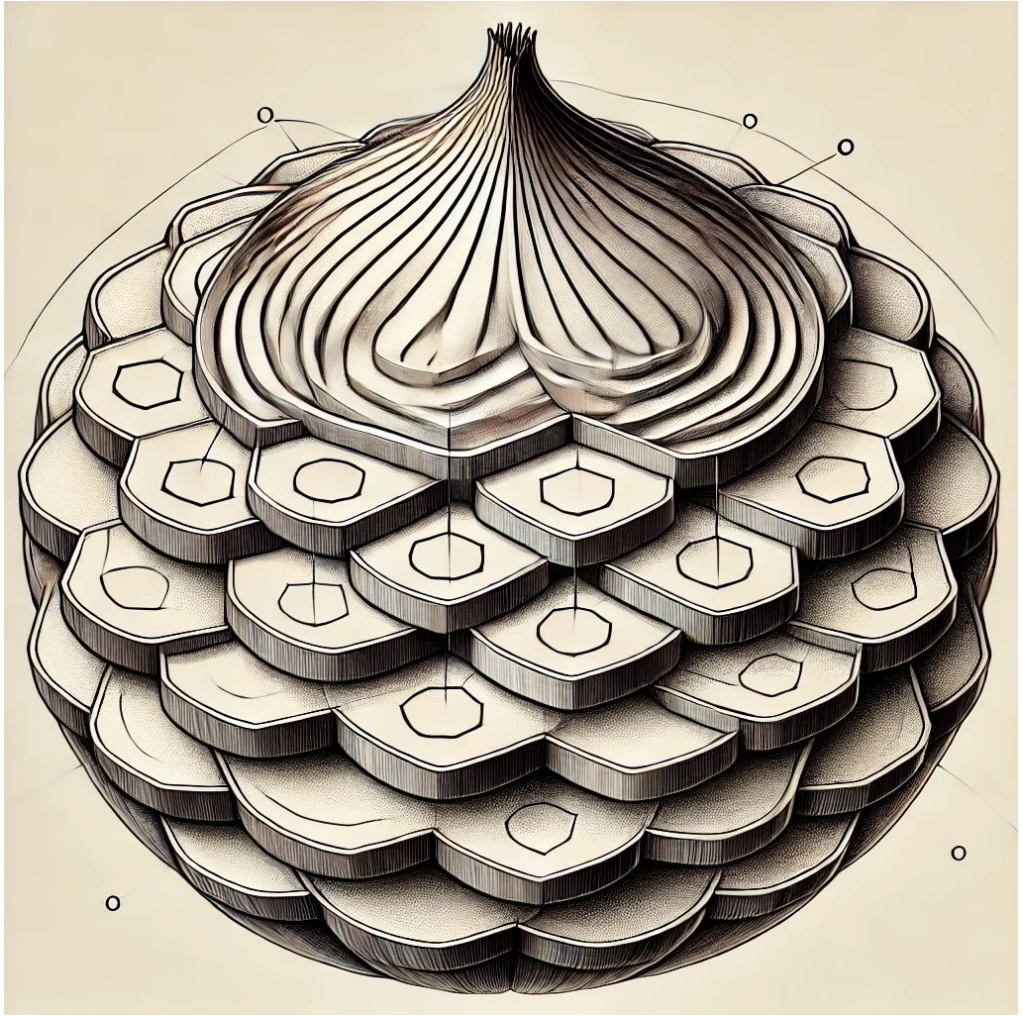
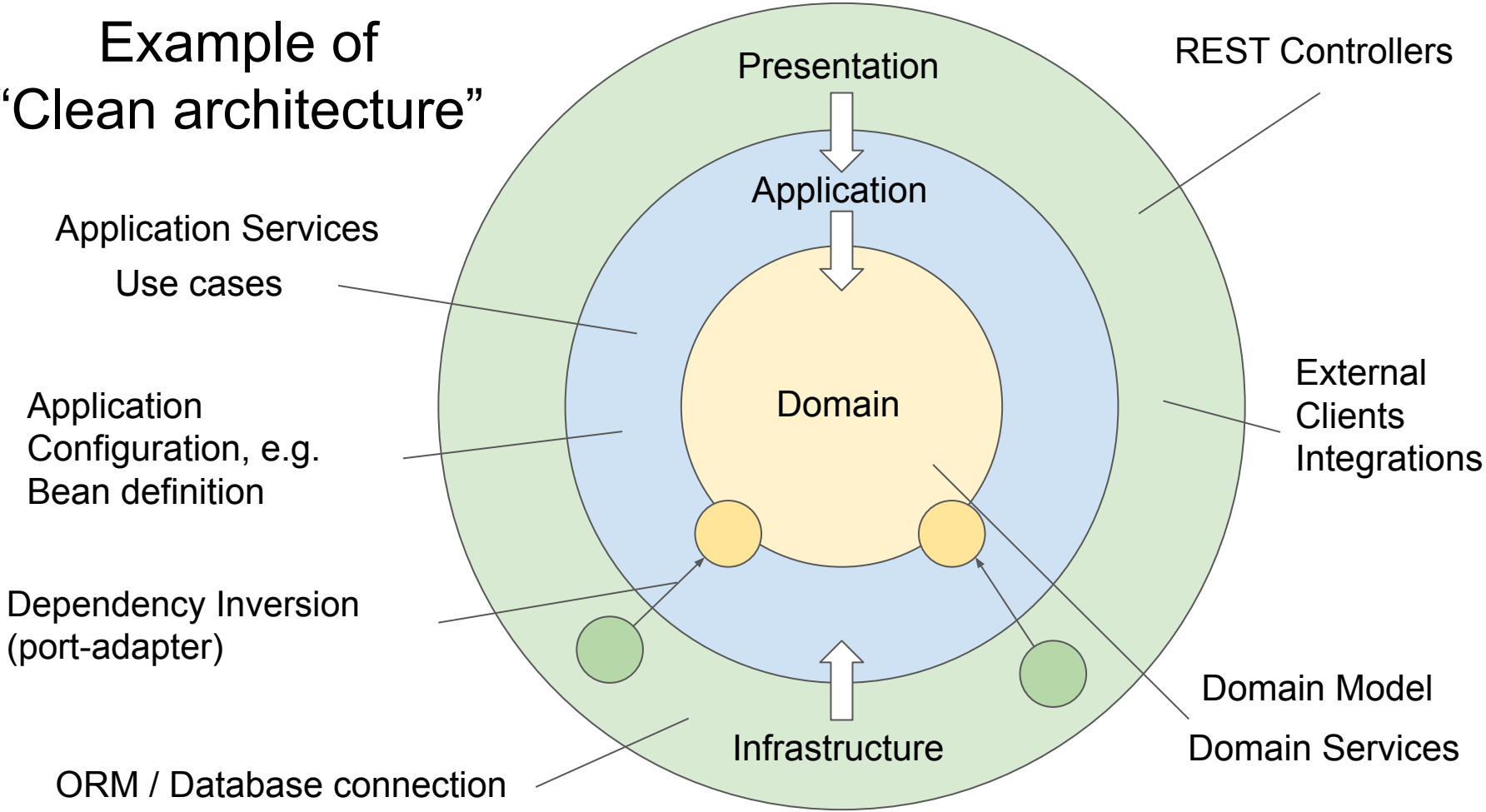


Image generated by AI using OpenAI's DALL-E

Example of “Clean architecture”



Refactoring toward “Clean Architecture”

Step 1: Product Listings

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POST /products

Add Product

GET /products

Product Listings

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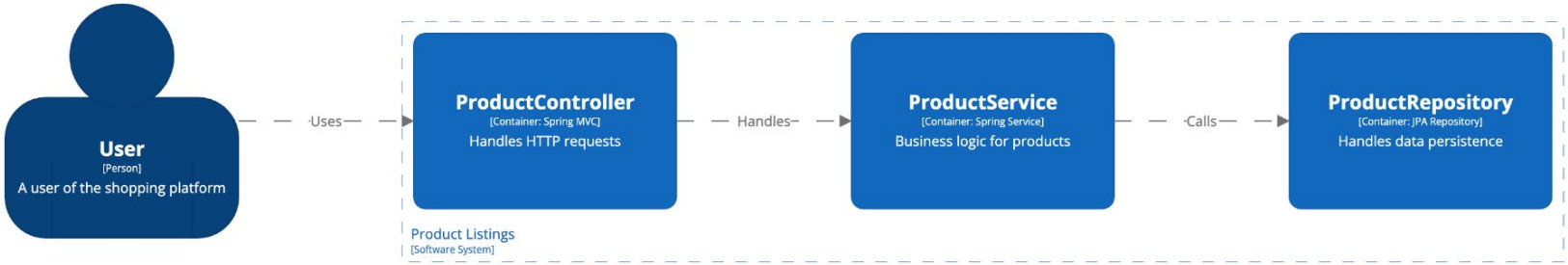

[Container] Product Listings
Thursday 19 September 2024 at 18:40 Central European Summer Time

Demo

Step 2: Manage Product Discounts

Two types of discounts:

- “N for the price of 1”
- Count-based percentage

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- “N for the price of 1”
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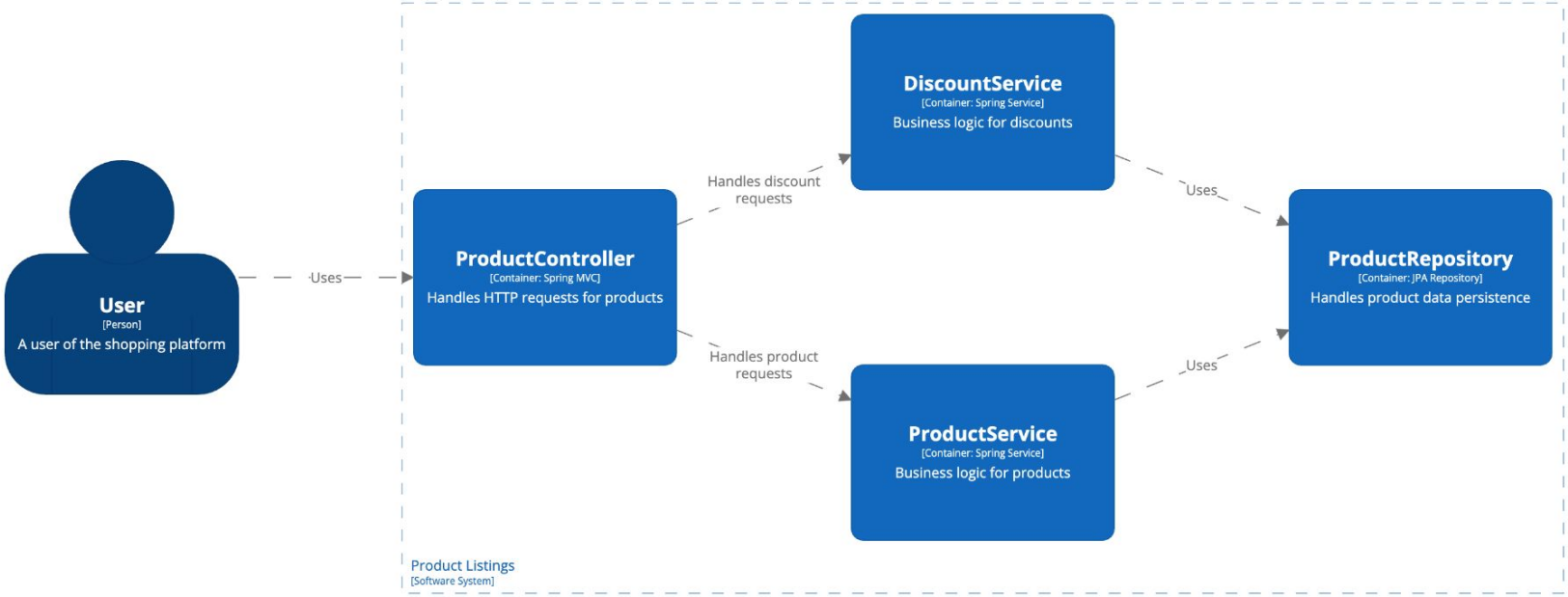
POST /products/{productId}/discounts

Add Discount To Product

POST /products/calculate-price

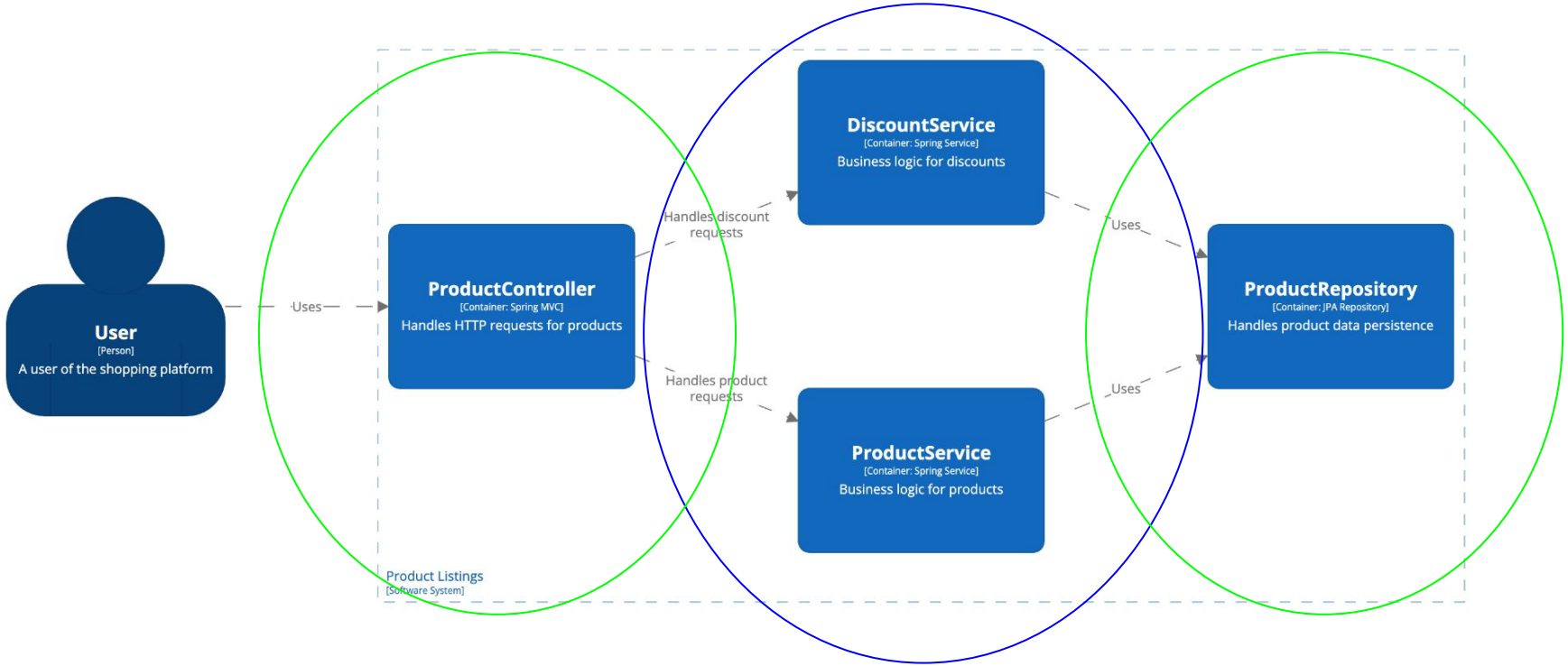
Products Pricing

Step 2: Manage Product Discounts



[Container] Product Listings
Thursday 19 September 2024 at 18:52 Central European Summer Time

Step 2: Manage Product Discounts



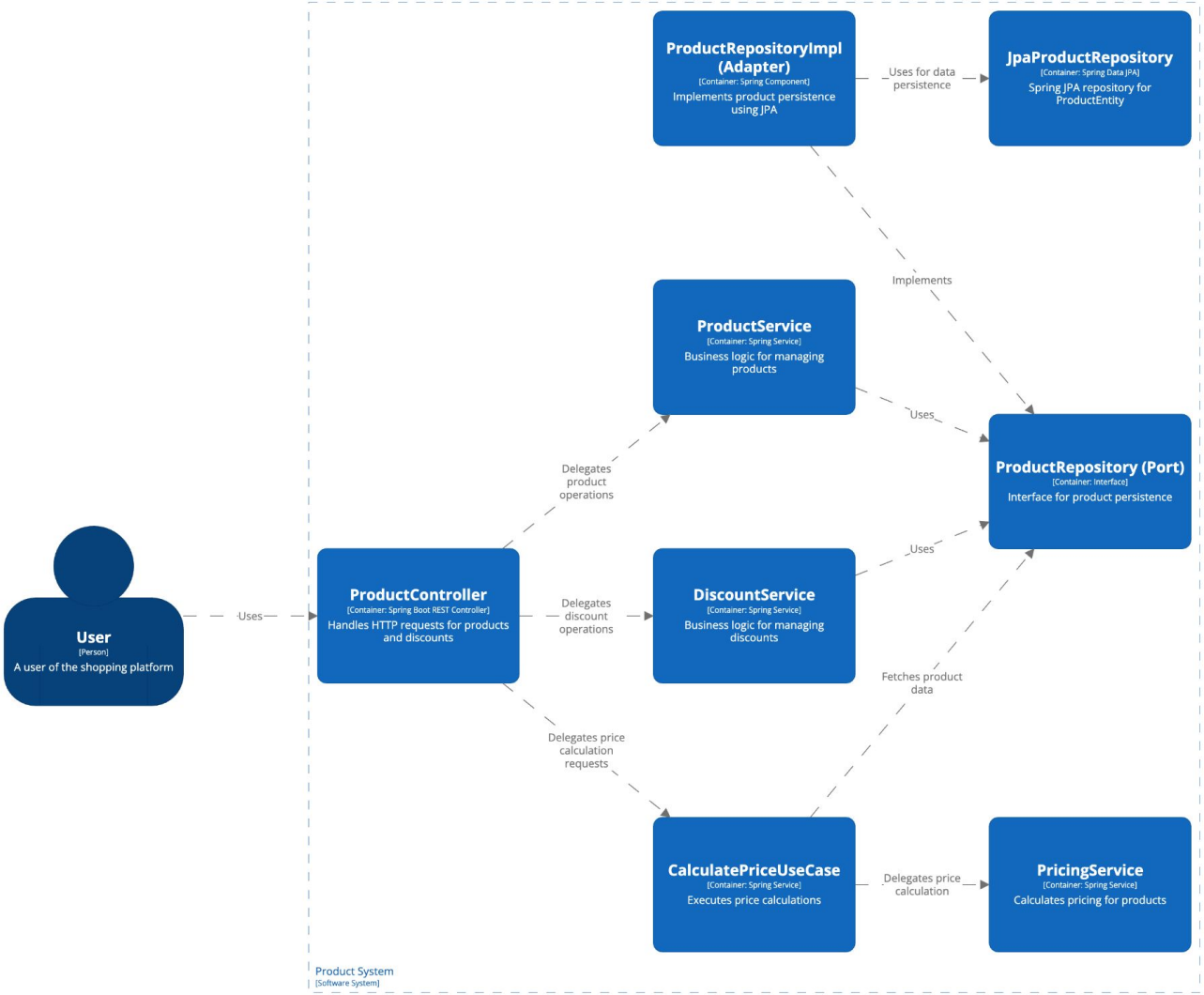
[Container] Product Listings
Thursday 19 September 2024 at 18:52 Central European Summer Time

Presentation layer

Business layer

Persistence layer

Demo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Spring JPA repository for ProductEntity
Implements
ProductService
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Business logic for managing products
Uses
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Interface for product persistence
Delegates product operations
Uses
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A user of the shopping platform
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Product System
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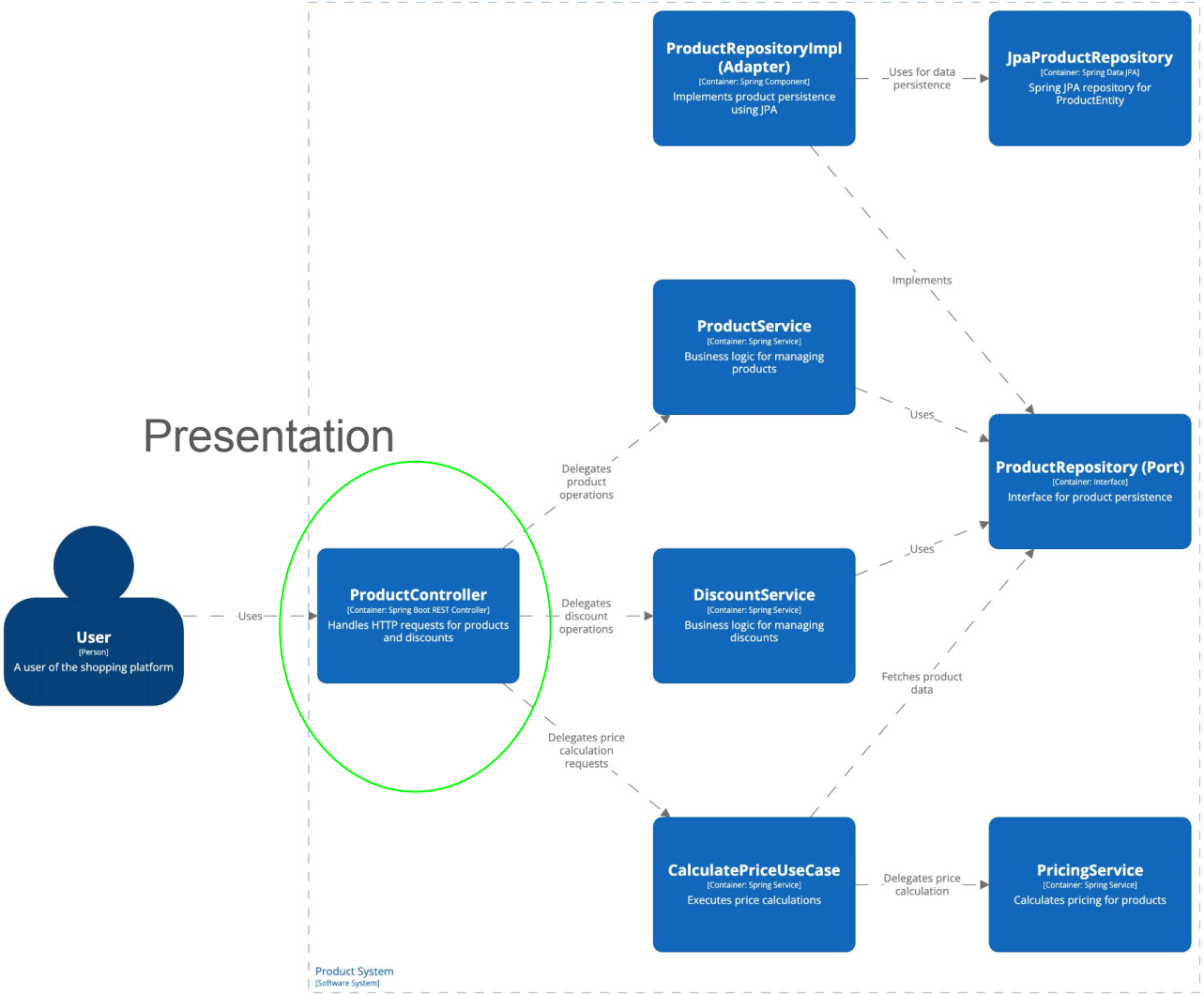
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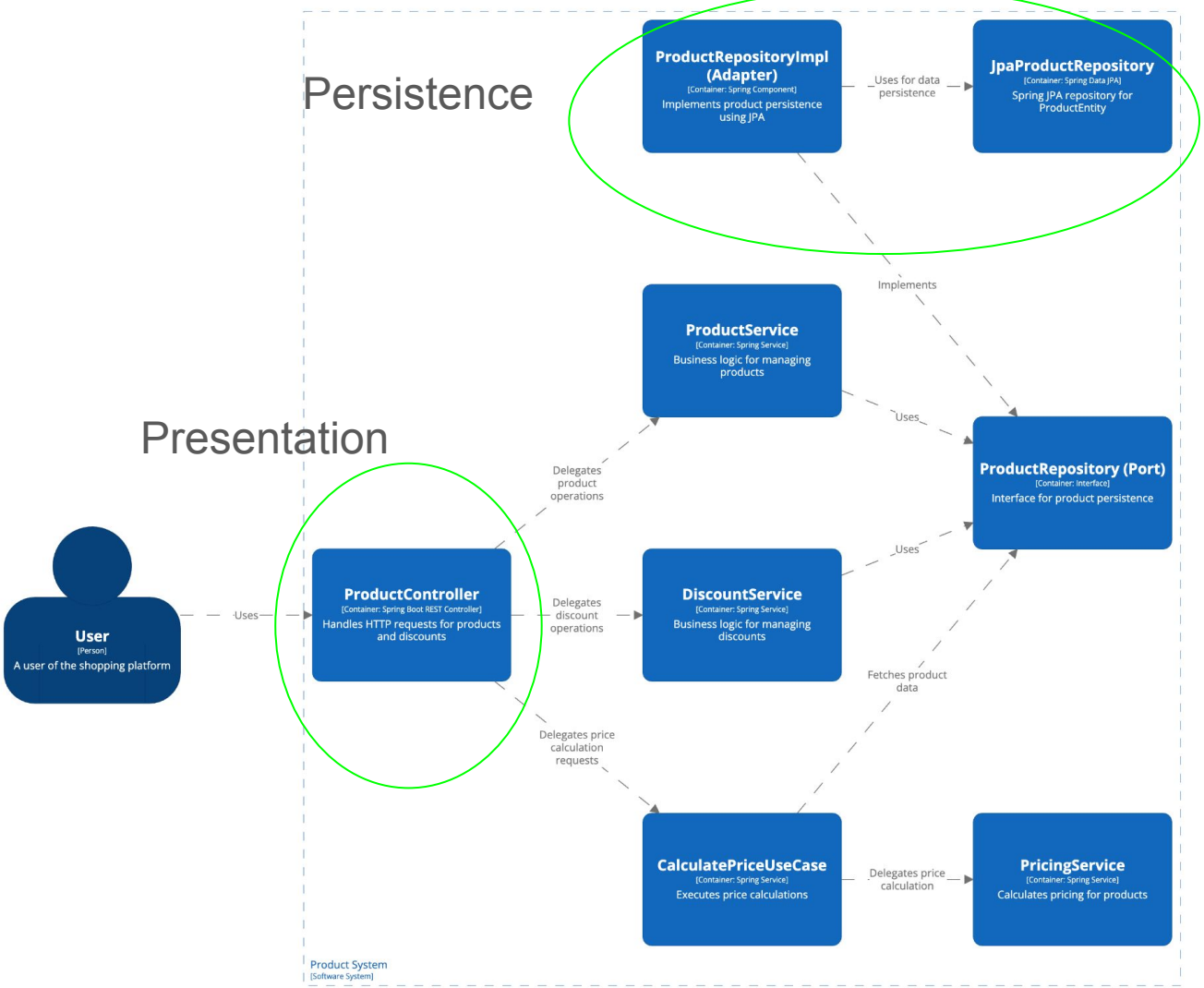
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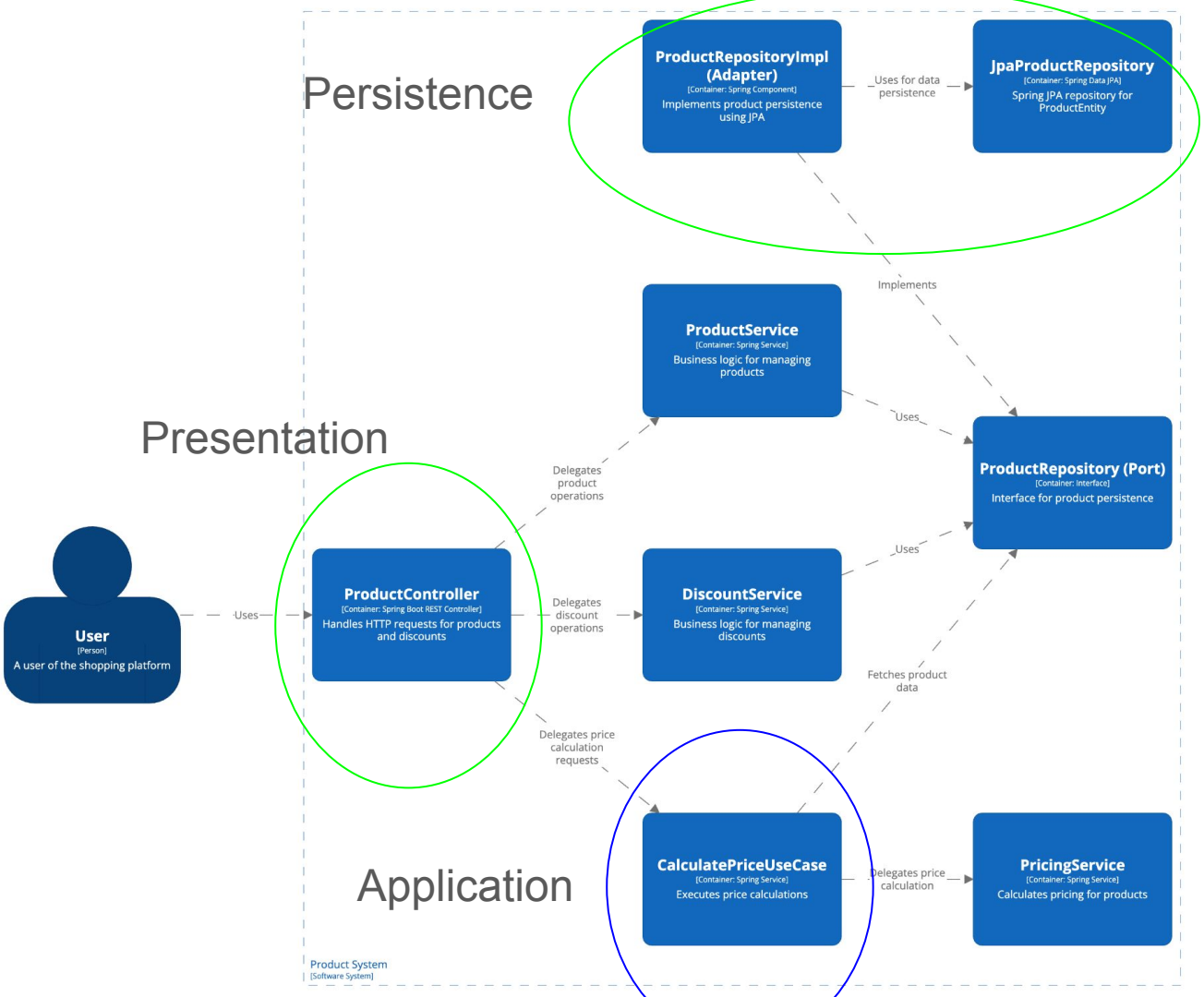
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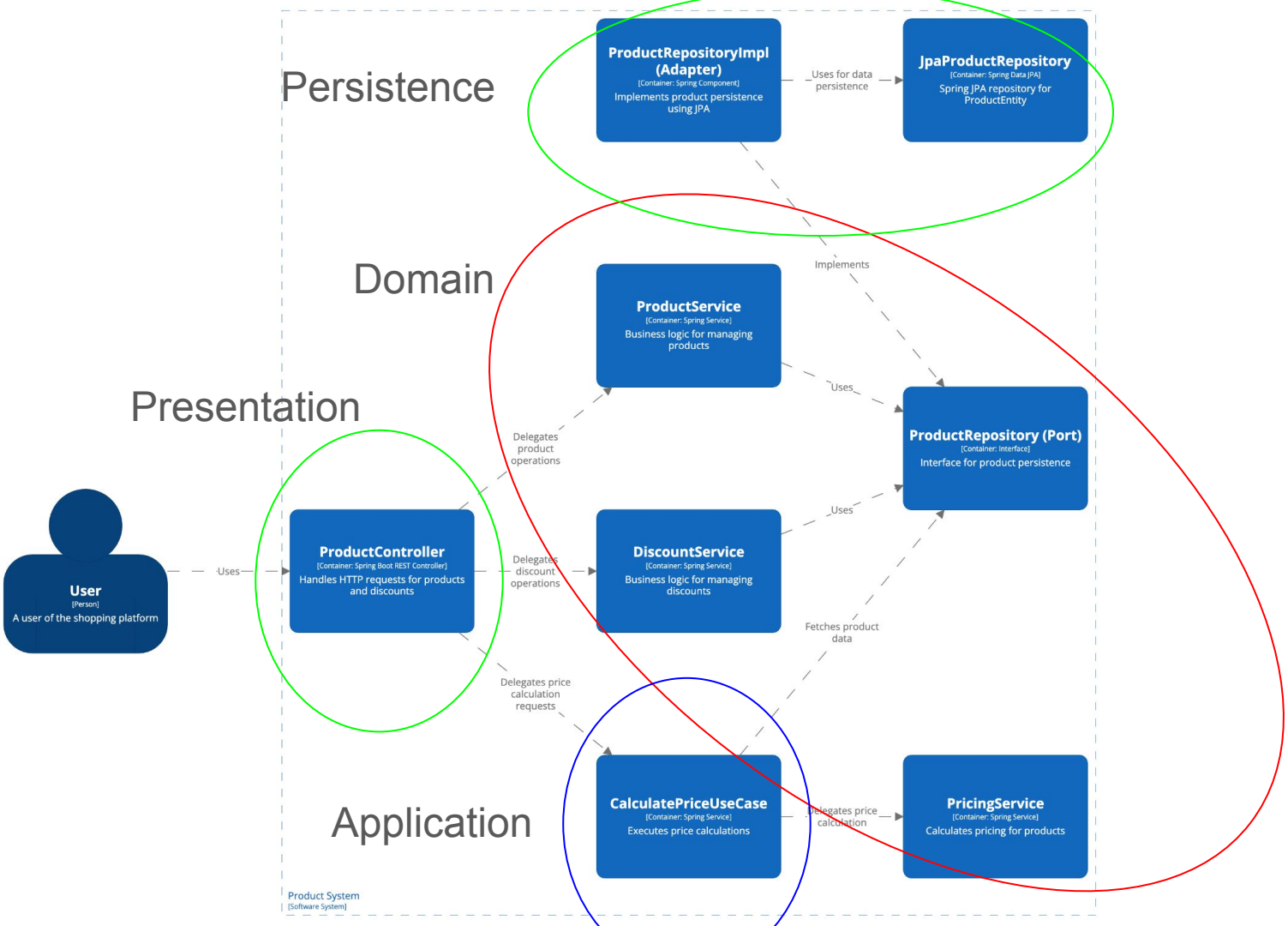
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- Mixed concerns,
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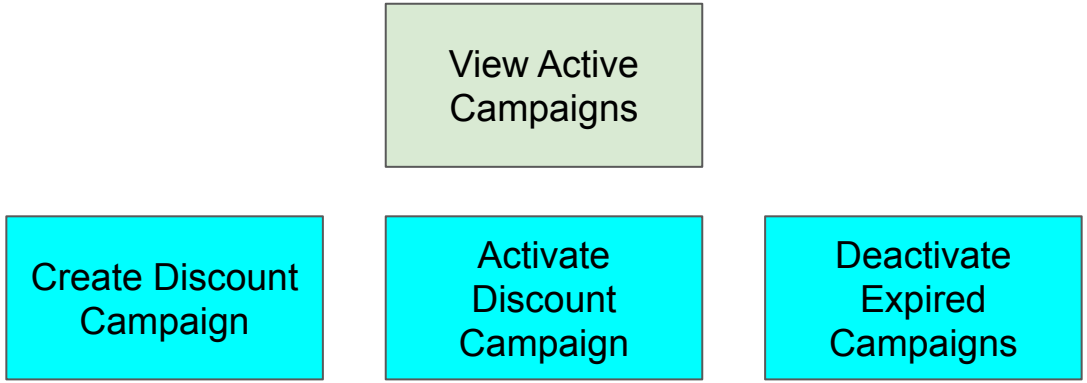
- Difficult to extend,
- Unclear structure - where to put my changes?

Step 3 - exercise: Discounts Campaigns

Business Requirement: Plan and Schedule Discount Campaigns



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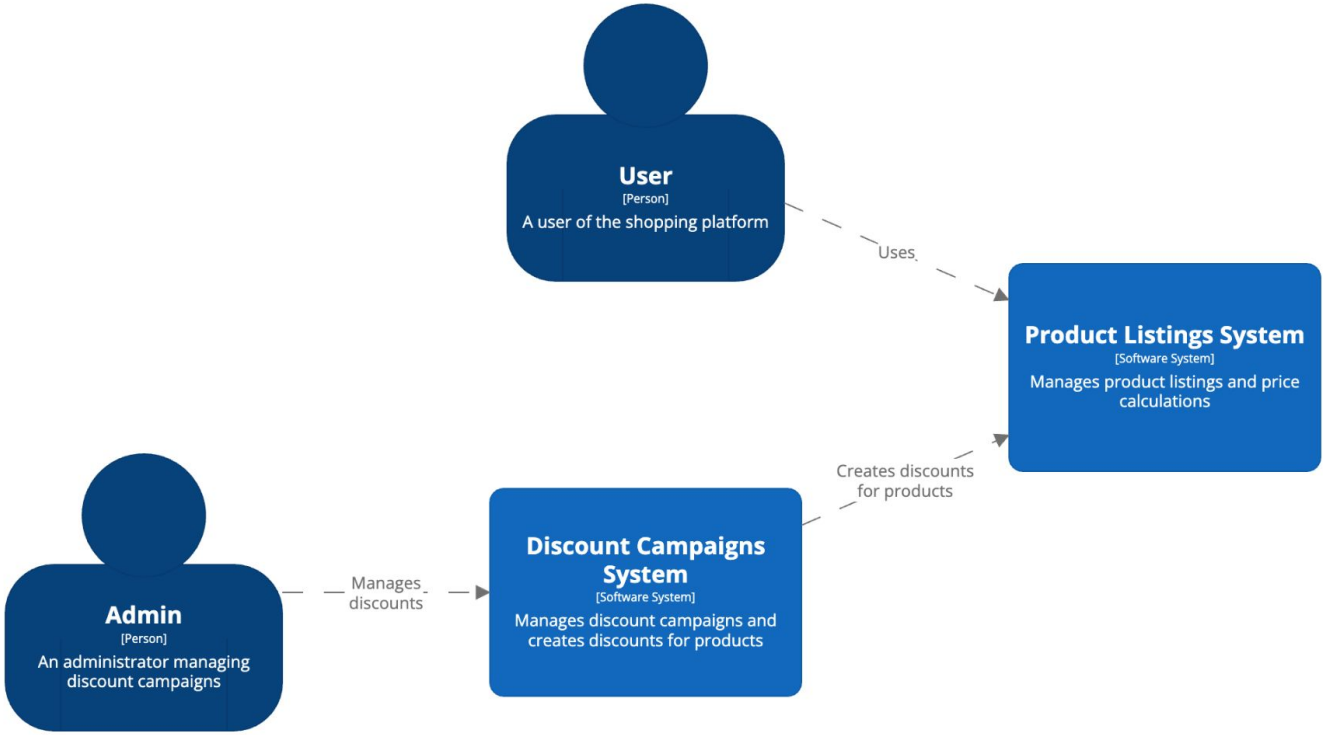

[System Context] Product Listings System

Monday 23 September 2024 at 08:19 Central European Summer Time

Additional Tips

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- Architectural drivers
  - ADRs might be helpful

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- KISS, SOLID, etc.
- Do not overengineer!

Q&A